Export to PDF: Word or pandoc; keep 11pt body.

H4 only — one page for meetings

Question: Within each outdoor bout, do heat, noise, light, pollution, and greenery relate to Empatica and eye-tracker signals after adjustment—and do **heat × sound** and **MRT × sound** match the hypothesis?

Input: `output/merged_all_11participants_8min.csv` · **Script:** `scripts/H4_multivariate.R` · **Outcomes:** 26 physiology series (no surveys).

Selection funnel (written before fits)

Step	In	Retained	Meaning
1	230	124	Tag every column; outcomes never become exposures
2	124	**16**	One representative per family (fixed rules)
3	16	**16**	Correlate; no VIF dropping
4	16	**6** designs	Pre-specified heat + PM layouts
5	—	26×6	Ten-second H4.1 LMMs

Figure: `output/statistical_analysis/results/H4_results/H4_Selection_Flow.pdf` · Register: `_work/H4_Multivariate_Filtration_Results.csv`.

16 exposures: air temp, RH, apparent wind, HUMIDEX, PET, UTCI, MRT, sun, dBA, PM1/2.5/10, lux, colour temp, GVI, SVF Holmer. Fit always adds exertion (`empatica_vector_magnitude`).

Six equations (shared non-thermal covariates + one PM slot each)

Eq	Role	Heat	PM
1	**Primary**	UTCI	PM2.5
2	Heat check	Temp + RH + wind	PM2.5
3	Heat check	HUMIDEX	PM2.5
4	Heat check	PET	PM2.5
5	PM check	UTCI	PM1
6	PM check	UTCI	PM10

Interaction rule

Each **heat term in that equation × `ucm_SND_dBA`**; always **`ucm_MRT × sound`**; no other exposure×sound terms. Phase: sum-to-zero among WalkG, WalkU, BikeG. Random: participant / phase nested; AR(1) on 10 s when supported.

FDR

Benjamini–Hochberg on **thermal main effects** only: one term in one equation across **26** outcomes. Heat×sound is reported, not the primary FDR family.

Open these first

`H4_Selection_Flow.pdf` · `_work/H4_Multivariate_Filtration_Results.csv` · `_work/H4_models/H4_coefficients.csv` · `H4_Report.md` · literature WHY: `Paper3_Github/Experiment papers/H1 to H4/H4_Screening_Literature_Guide.md`

More detail: `Paper3_H4_How_We_Did_It.md` · whole study: `scripts/Paper3_Methods_Plain.md`.